

OVERVIEW

I am an assistant professor of Statistics at Texas A&M University. My research focuses on computational optimal transport, Bayesian statistics, and statistical machine learning.

EDUCATION

The University of Texas at Austin Ph.D. in Statistics	Texas, USA 2021–2026
Hanoi University of Science and Technology (HUST) B.Sc in Computer Science (5 years program)	Hanoi, Vietnam 2015–2020

ACADEMIC POSITIONS

Texas A&M University <i>Assistant Professor of Statistics</i>	Texas, USA August, 2026 –Now
The University of Texas at Austin <i>Graduate Research Assistant and Graduate Teaching Assistant</i>	Texas, USA September, 2022 –May, 2025

INDUSTRIAL EXPERIENCE

Amazon Science <i>Applied Scientist Intern</i>	Austin, TX, USA May, 2025 –August, 2025
– Developed a multi-task training framework session-based recommendations with large language models and improved online-inference by using KV caching and speculative decoding.	
Amazon Science <i>Applied Scientist Intern</i>	Seattle, WA, USA May, 2024 –August, 2024
– Proposed a framework combining large language models and deep generative models to enhance product recommendation diversity.	
Toyota InfoTech Labs <i>Research Intern</i>	Mountain View, CA, USA May, 2023 –August, 2023
– Developed a Transformer-based framework with global-local decomposition for improved battery health prediction, resulting in patent US12430962B2.	
AT&T Labs <i>Research Intern</i>	Texas, USA June, 2022 –August, 2022
– Proposed and implemented co-clustering algorithms to analyze user browsing behavior in PySpark.	
VinAI Research (Acquired by Qualcomm AI Research) <i>AI Research Resident</i>	Hanoi, Vietnam 2019 –2021
– Did research on Deep Generative Models (VAEs, GANs, score matching, diffusion models) and improved them with Optimal Transport (sliced Wasserstein distance, Sinkhorn divergence).	

AWARDS

- **Stellar Abstract Award at STAI-X.** 2026
- **Bartroff Family Award for Research Excellence.** 2026
- **Schmidt Science Fellows Finalist.** 2026
- **UT Austin Professional Development Award.** 2026
- **Top Reviewer Award at NeurIPS.** 2025
- **BNP14 Travel Award.** 2025
- **UT Austin Outstanding Graduate Research Fellowship.** 2025
The UT Austin Outstanding Graduate Research Fellowship is one of the university's most prestigious awards, recognizing exceptional academic achievement and research excellence among continuing graduate students.
- **Top Reviewer Award at NeurIPS.** 2024
- **ICML Travel Grants.** 2023
- **Top Reviewer Award at NeurIPS.** 2022
- **NeurIPS Scholar Award.** 2022
- **ICML Travel Grants.** 2022
- **UT Austin Graduate Recruitment Fellowship.** 2021
The UT Austin Graduate Recruitment Fellowship is a highly prestigious award offered to attract top academic talents.

BOOKS

1. **K. Nguyen**, “An Introduction to Sliced Optimal Transport: Foundations, Advances, Extensions, and Applications”, *Foundations and Trends® in Computer Graphics and Vision*, 2025.
2. BDL Consortium, “Bayesian deep learning”, *CRC Press*, 2026, David Dunson, Vincent Fortuin, Yarin Gal, Philipp Hennig, Konstantina Palla, Theodore Papamarkou, Tim Rudner, Sara Wade, Andrew Wilson, Cheng Zhang, and **Khai Nguyen** among other contributing authors; under preparation.

REFEREED PUBLICATIONS

(*) denotes equal contribution,

Google Scholar: <https://scholar.google.com/citations?user=im5fNaQAAAAJ&hl=en>

1. **K. Nguyen**, “Streaming sliced optimal transport”, *International Conference on Machine Learning (ICML)*, 2026.
2. **K. Nguyen***, H. Nguyen*, T. Pham, and N. Ho, “Fast estimation of Wasserstein distances via regression on sliced Wasserstein distances”, *International Conference on Learning Representations (ICLR)*, 2026.
3. **K. Nguyen** and P. Mueller, “Summarizing nonparametric Bayesian mixture posterior – sliced optimal transport metrics for Gaussian mixtures”, *Journal of Computational and Graphical Statistics (JCGS)*, 2026.
4. M. Luong, **K. Nguyen**, D. Phung, G. Haffari, and L. Qu, “Unbiased sliced Wasserstein kernels for high-quality audio captioning”, *Neural Information Processing Systems (NeurIPS)*, 2025.
5. **K. Nguyen***, H. Nguyen*, T. Pham, and N. Ho, “Lightspeed geometric dataset distances via sliced optimal transport”, *International Conference on Machine Learning (ICML)*, 2025.

6. **K. Nguyen***, H. Nguyen*, and N. Ho, “Towards marginal fairness sliced Wasserstein barycenter”, *International Conference on Learning Representations (ICLR)*, **Spotlight 3.2%**, 2025.
7. **K. Nguyen** and N. Ho, “Hierarchical hybrid sliced Wasserstein: A scalable metric for heterogeneous joint distributions”, *Neural Information Processing Systems (NeurIPS)*, 2024.
8. **K. Nguyen**, S. Zhang, T. Le, and N. Ho, “Sliced Wasserstein with random-path projecting directions”, *International Conference on Machine Learning (ICML)*, 2024.
9. T. T. Le, **K. Nguyen**, S. Sun, N. Ho, and X. Xie, “Integrating efficient optimal transport and functional maps for unsupervised shape correspondence learning”, in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
10. **K. Nguyen**, N. Barileto, and N. Ho, “Quasi-Monte Carlo for 3D sliced Wasserstein”, *International Conference on Learning Representations (ICLR)*, **Spotlight 5%**, 2024.
11. **K. Nguyen** and N. Ho, “Sliced Wasserstein estimation with control variates”, *International Conference on Learning Representations (ICLR)*, 2024.
12. T. Le, **K. Nguyen**, N. Ho, S. Sun, K. Han, and X. Xie, “Diffeomorphic deformation via sliced Wasserstein distance optimization for cortical surface reconstruction”, *International Conference on Learning Representations (ICLR)*, 2024.
13. M. Luong, **K. Nguyen**, N. Ho, R. Haf, D. Phung, and L. Qu, “Revisiting deep audio-text retrieval through the lens of transportation”, *International Conference on Learning Representations (ICLR)*, 2024.
14. H. Nguyen, **K. Nguyen**, and N. Ho, “On parameter estimation in deviated Gaussian mixture of experts”, *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
15. H. Nguyen, T. Nguyen, **K. Nguyen**, and N. Ho, “Towards convergence rates for parameter estimation in Gaussian-gated mixture of experts”, *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
16. **K. Nguyen** and N. Ho, “Energy-based sliced Wasserstein distance”, *Neural Information Processing Systems (NeurIPS)*, 2023.
17. **K. Nguyen**, T. Ren, and N. Ho, “Markovian sliced Wasserstein distances: Beyond independent projections”, *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
18. D. Le*, H. Nguyen*, **K. Nguyen***, T. Nguyen, and N. Ho, “Fast approximation of the generalized sliced-Wasserstein distance”, *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
19. X. Han, T. Ren, T. M. Nguyen, **K. Nguyen**, J. Ghosh, and N. Ho, “Robustify Transformers with robust kernel density estimation”, *Neural Information Processing Systems (NeurIPS)*, 2023.
20. D. Do, H. Nguyen, **K. Nguyen**, and N. Ho, “Minimax optimal rate for parameter estimation in multivariate deviated models”, *Neural Information Processing Systems (NeurIPS)*, 2023.
21. **K. Nguyen***, D. Nguyen*, and N. Ho, “Self-attention amortized distributional projection optimization for sliced Wasserstein point-clouds reconstruction”, *International Conference on Machine Learning (ICML)*, 2023.
22. **K. Nguyen**, T. Ren, H. Nguyen, L. Rout, T. Nguyen, and N. Ho, “Hierarchical sliced Wasserstein distance”, *International Conference on Learning Representations (ICLR)*, 2023.

23. D. Nguyen, T. Nguyen, **K. Nguyen**, D. Phung, H. Bui, and N. Ho, “On cross-layer alignment for model fusion of heterogeneous neural networks”, *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
24. **K. Nguyen** and N. Ho, “Revisiting sliced Wasserstein on images: From vectorization to convolution”, *Neural Information Processing Systems (NeurIPS)*, 2022.
25. **K. Nguyen** and N. Ho, “Amortized projection optimization for sliced Wasserstein generative models”, *Neural Information Processing Systems (NeurIPS)*, 2022.
26. T. Nguyen, M. Pham, T. Nguyen, **K. Nguyen**, S. J. Osher, and N. Ho, “Transformer with Fourier integral attentions”, *Neural Information Processing Systems (NeurIPS)*, 2022.
27. T. Nguyen, T. Nguyen, H. Do, **K. Nguyen**, V. Saragadam, M. Pham, K. Nguyen, N. Ho, and S. J. Osher, “Improving transformer with an admixture of attention heads”, *Neural Information Processing Systems (NeurIPS)*, 2022.
28. **K. Nguyen**^{*}, D. Nguyen^{*}, T.-A. Vu-Le, T. Pham, and N. Ho, “Improving mini-batch optimal transport via partial transportation”, in *International Conference on Machine Learning (ICML)*, 2022.
29. **K. Nguyen**, D. Nguyen, Q. Nguyen, T. Pham, H. Bui, D. Phung, T. Le, and N. Ho, “On transportation of mini-batches: A hierarchical approach”, in *International Conference on Machine Learning (ICML)*, 2022.
30. K. Le, H. Nguyen, **K. Nguyen**, T. Pham, and N. Ho, “On multimarginal partial optimal transport: Equivalent forms and computational complexity”, in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, 2022, pp. 4397–4413.
31. S. Nguyen, D. Nguyen, **K. Nguyen**, K. Than, H. Bui, and N. Ho, “Structured dropout variational inference for bayesian neural networks”, *Neural Information Processing Systems (NeurIPS)*, 2021.
32. **K. Nguyen**, S. Nguyen, N. Ho, T. Pham, and H. Bui, “Improving relational regularized autoencoders with spherical sliced fused Gromov-Wasserstein”, in *International Conference on Learning Representations (ICLR)*, 2021.
33. **K. Nguyen**, N. Ho, T. Pham, and H. Bui, “Distributional sliced-Wasserstein and applications to generative modeling”, in *International Conference on Learning Representations (ICLR)*, **Spotlight 3.78%**, 2021.

WORKS UNDER REVIEW, REVISION, AND PREPARATION

1. **K. Nguyen**, Y. Ni, E. Juarez-Colunga, and P. Mueller, “Distributional determinantal point process and repulsive clustering of distributions”, *Under review*, 2026.
2. **K. Nguyen**, E. Juarez-Colunga, and P. Mueller, “Generalized Bayesian clustering with regression for unaligned longitudinal binary data”, *Under Review*, 2026.
3. **K. Nguyen**, “Sliced-regularized optimal transport”, *Under review*, 2026.
4. M.-P. Truong and **K. Nguyen**, “Amortized optimal transport from sliced potentials”, *Under review*, 2026.
5. **K. Nguyen**, Y. Ni, and P. Mueller, “Vertical consensus inference for high-dimensional random partition”, *Under review*, **Stellar Abstract Award at STAI-X**, 2026.

6. **K. Nguyen**, Y. Ni, and P. Mueller, “Bayesian multiple multivariate density-density regression”, *Under review*, 2026.
7. **K. Nguyen**, Y. Ni, and P. Mueller, “Bayesian multivariate density-density regression”, *Under major revision at Bayesian Analysis*, 2025.

OTHER WORKS

1. **K. Nguyen**, “Discussion of “Robust probabilistic inference via a constrained transport metric””, *Bayesian Analysis (Contributed Discussion)*, 2026.
2. C. Wang, **K. Nguyen**, and A. T. Pham, *Battery life prediction using a global-local decomposition transformer*, US Patent 12,430,962, Sep. 2025.

TALKS

- Statistics and Trustworthy AI for Cross (X)-Domain Acceleration (Stellar Abstract Award Talk): “Vertical consensus inference for high-dimensional random partition”, 2026.
- Conference of Texas Statisticians (Contributed Talk): “Bayesian Multiple Multivariate Density-Density Regression”, 2026.
- Texas A&M University, Department of Statistics (Colloquium): “Sliced Optimal Transport and Statistics”, 2026.
- The Ohio State University, Department of Statistics (Colloquium): “Sliced Optimal Transport and Statistics”, 2026.
- The Institute for Computational and Experimental Research in Mathematics (ICERM), Nonparametric Bayesian Inference - Computational Issues (Lightning Talk): “Bayesian Multiple Multivariate Density-Density Regression”, 2026.
- Florida State University, Department of Statistics (Colloquium): “Sliced Optimal Transport and Statistics”, 2025.
- The Bayesian Young Statisticians Meeting (Talk in session with discussion): “Summarizing Bayesian Nonparametric Mixture Posterior - Sliced Optimal Transport Metrics for Gaussian Mixtures”, 2025.
- International Conference on Bayesian Nonparametrics (Contributed Talk): “Summarizing Bayesian Nonparametric Mixture Posterior - Sliced Optimal Transport Metrics for Gaussian Mixtures”, 2025.
- International Conference on Machine Learning (ICML) (Spotlight Talk): “On Transportation of Mini-batches: A Hierarchical Approach”, 2022.
- International Conference on Machine Learning (ICML) (Spotlight Talk): “Improving Mini-batch Optimal Transport via Partial Transportation”, 2022.
- International Conference on Learning Representation (ICLR) (Spotlight Talk): “Distributional Sliced-Wasserstein and Applications to Generative Modeling”, 2021.

PROFESSIONAL SERVICES

- Reviewer at Journal of Machine Learning Research (JMLR).
- Reviewer at Journal of Computational and Graphical Statistics (JCGS).
- Reviewer at Bayesian Analysis.
- Reviewer at Statistics and Computing.
- Reviewer at IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI).
- Reviewer at Transactions on Machine Learning Research (TMLR).
- Reviewer at IEEE Transactions on Information Theory.
- Reviewer at Machine Learning Journal.
- Reviewer at International Conference on Machine Learning (ICML) 2022-2026.

- Reviewer at Conference on Neural Information Processing Systems (NeurIPS) 2021-2025.
- Reviewer at International Conference on Learning Representations (ICLR) 2022-2026.
- Reviewer at International Conference on Artificial Intelligence and Statistics (AISTATS) 2022-2025.
- Reviewer at Statistics and Trustworthy AI for Cross (X)-Domain Acceleration (STAI-X) 2026.
- Reviewer at AAAI Conference on Artificial Intelligence (AAAI) 2023-2026 .
- Reviewer at IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR) 2023-2026.
- Reviewer at International Conference on Computer Vision (ICCV) 2023.
- Reviewer at European Conference on Computer Vision (ECCV) 2024.
- Reviewer at IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2026.
- Reviewer at IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2024-2025.
- Reviewer at Conference on Language Modeling (COLM) 2024-2025.
- Reviewer at Learning on Graphs Conference (LOG) 2024-2025.

MENTORSHIP

- Minh-Phuc Truong, 2025-Now, Undergraduate student at HUST.
- Truong Nguyen, 2025-Now, Undergraduate student at HUST.
- Tien-Phat Nguyen, 2025-Now, Undergraduate student at HUST.
- Hai Nguyen, 2024-2026, Undergraduate student at HUST, Now: Ph.D Student at Cornell ECE

PROFESSIONAL MEMBERSHIP

- American Statistical Association (ASA).
- International Society for Bayesian Analysis (ISBA).
- Institute of Mathematical Statistics (IMS).
- Society for Industrial and Applied Mathematics (SIAM).
- Institute of Electrical and Electronics Engineers (IEEE).